

Route Planning (TSP)

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Mode: Full Pipeline

1. Problem Formulation

Problem Type:	Find
Domain:	Mathematical Optimization
Named Problem:	Traveling Salesman Problem

Universe

- See problem description

Variables

Symbol	Meaning	Type / Range
x	decision variable	$\mathbb{R}$

Structure

A delivery driver must visit 8 locations in a city and return to the warehouse. Given distances between all pairs of locations, find the shortest possible route that visits every location exactly once and returns to the starting point.

Real-World Mapping

Real-World Concept	Mathematical Object
total score/cost	objective function value

Constraints

(1) Traveling Salesman Problem (TSP)	classic NP-hard combinatorial optimization
(2) Held-Karp DP	exact algorithm using bitmask dynamic programming
(3) Nearest-neighbor heuristic	greedy baseline for comparison
(4) Tour verification	independent check that solution visits all cities exactly once

Objective

optimize $f(x)$

Approach

Suggested approachHeld-Karp DP (exact, $O(n^2 * 2^n)$, $n=9$)

Complexity class: $O(n^2 * 2^n)$

Available tools: Held-Karp DP

2. Solution

Answer:	Solution computed
Objective Value:	117
Optimal:	Yes
Feasible:	Yes
Algorithm:	Held-Karp DP (exact, $O(n^2 * 2^n)$, $n=9$) $O(n^2 * 2^n)$
Time:	0.0013s
Certificate:	Solver reports optimal.

Solution Details

Objective value: $z^* = 117$

Method: Held-Karp dynamic programming for exact TSP solution.

- Time complexity: $O(n^2 * 2^n)$ -- feasible for $n=9$
- Space complexity: $O(n * 2^n)$
- Guaranteed optimal solution (exact algorithm)
- Compared against nearest-neighbor greedy heuristic for context

Verification

✓ Feasibility	All constraints satisfied
✓ Optimality	Held-Karp DP (exact, $O(n^2 * 2^n)$, $n=9$)

3. Interpretation

The Question

A delivery driver must visit 8 locations in a city and return to the warehouse.

The Answer

The optimal solution has objective value $z^* = 117$.

What This Means

- Optimal tour visiting all 9 locations (warehouse + 8 stops)
- Total tour distance
- Step-by-step route with individual leg distances
- Comparison with nearest-neighbor heuristic
- Independent verification of tour validity and distance

Held-Karp dynamic programming for exact TSP solution.

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Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution